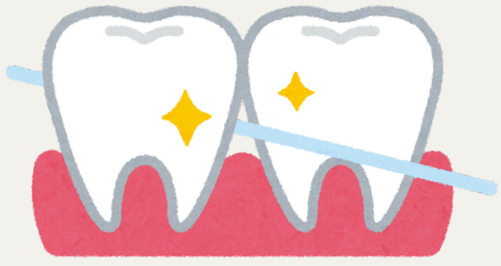





All Ceramic Crown Prep




A Full Guide




What is an All Ceramic Crown Prep?



An all ceramic crown prep is the type of prep that is done to place a **full coverage crown** made of a **ceramic** material.
This material could be porcelain, zirconia or Emax.

It is usually done in case of:

- Crown is planned in the esthetic zone (usually from upper premolar to the upper premolar on the other side.
- Patient has high esthetic demands (even outside of the esthetic zone)





The picture shows full coverage ceramic crowns planned in the esthetic zone (from right upper 4 to left upper 4)





Preparation Criteria



- The total reduction needed for all ceramic crown prep is 1 mm all around.
- The incisal edge should be reduced by 1.5-2mm.
- The lingual fossa should be reduced by 1 mm.
- Be aware that the reduction of the lingual surface is separate from the reduction of the lingual fossa.(will be explained further in coming slides)
- Finish line should have a thickness of 1 mm.
- Margin should be rounded with no sharp internal angles.
- Finish line type should be Deep Chamfer or Rounded Shoulder. Deep Chamfer will be the type used in the coming video demos.

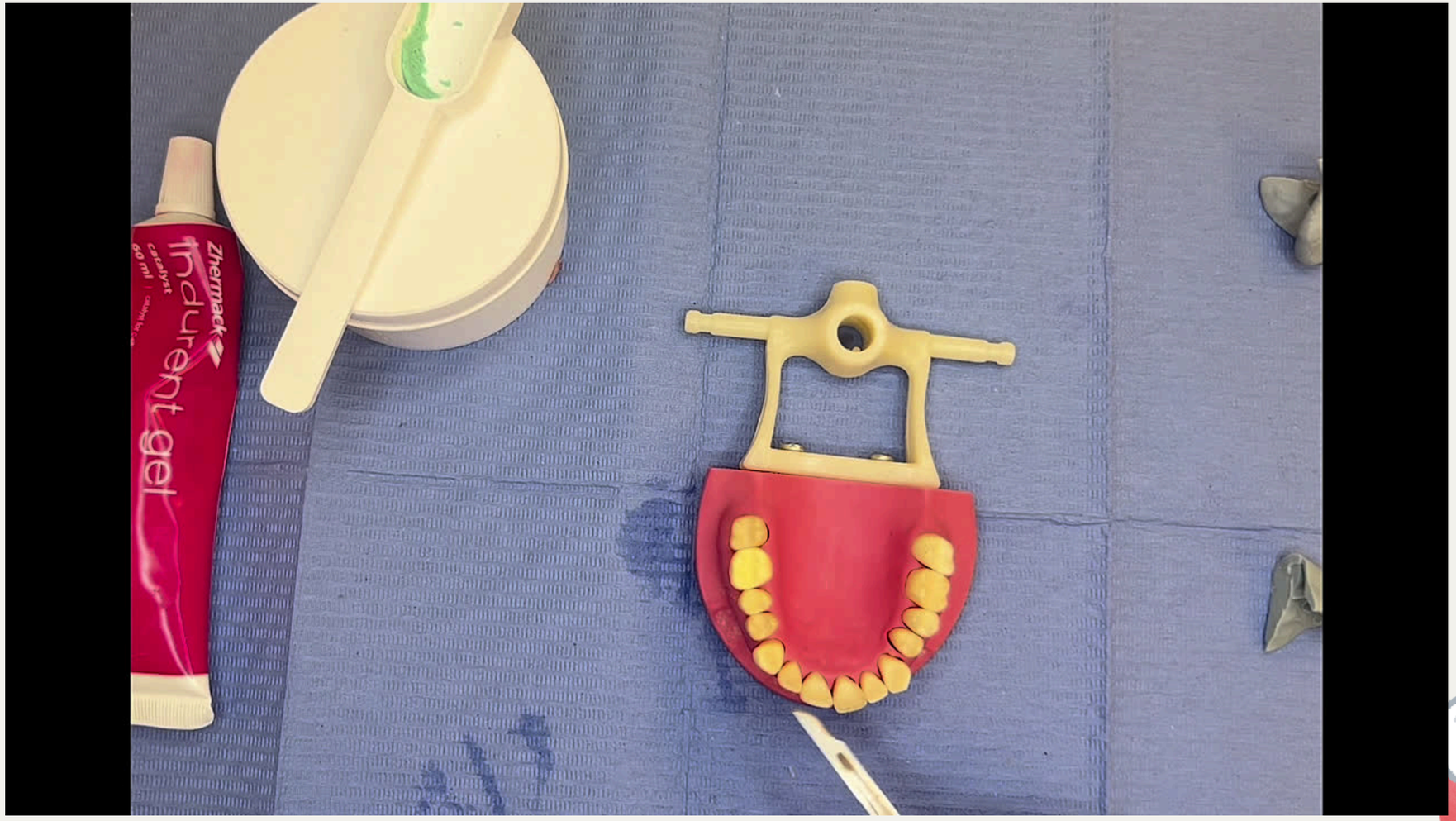




Measuring the amount of reduction



To measure the amount of reduction accurately, use a condensation silicon impression using the technique demonstrated in the video below:









What you need for the perfect prep





Micromotor for low speed



High/low speed Contra



Mirror+ explorer



Probe



Tapered with round end blue



Needle blue



Flame/football blue



Flame/football yellow



Tapered with round yellow



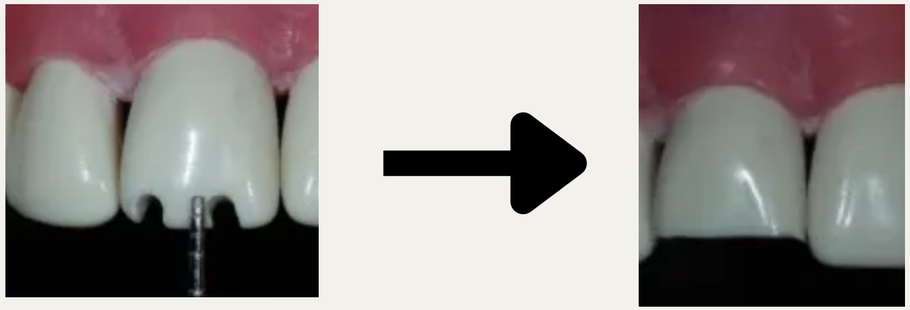
The Incisal Edge



The incisal edge should be reduced by 1.5- 2mm to leave enough space for the ceramic material to be able to withstand the occlusal forces.
When reducing the incisal edge, there are two methods you can follow:

- Depth Grooves method: using a tapered with round bur create grooves that correspond to depth of 1.5- 2mm.

How do you know that you reached the correct depth? Use HALF the thickness of the stone to create the grooves then measure it with a probe. Next remove the tooth structure in between the grooves. Now you have an evenly reduced incisal edge! Easy peasy!





- Wheel Stone method: NOT begginner friendly! Wheel stone removes tooth structure very quickly.

Use a wheel stone for a quick and smooth reduction of incisal edge . This method needs a trained eye to know when to stop. Measure regularly using your silicone impression to be sure.

The Incisal Edge





Axial Reduction



- The required amount of reduction is 1mm.
- The finish line thickness should be 1mm.
- The finish line type is Deep Chamfer.

Don’t forget that reduction is done on 2 planes on the facial surface:

- Primary plane: parallel to the long axis of the tooth.
- Secondary plane: done at 45 degrees.

<u>The Palatal Surface has a few considerations that are specific to it:</u>

- The palatal surface reduction is separate from lingual fossa reduction.
- Use a flame/football blue stone for easy reduction of the lingual fossa.
- While reducing the fossa, do not touch the remaining parts of the palatal surface outside the fossa.
- After reduction is complete, the cingulum should still be detectable, not completely lost.





Axial Reduction



Axial Reduction





Proximal Reduction



What you need to know:

- When reducing a proximal surface (Mesial/Distal), you need to use a **Needle Stone** to open the **Contact** with the **adjacent tooth.**
- Keep using the needle stone till there is enough space for Tapered with Round End to pass **smoothly.**
- **NEVER** use a needle stone to create your finish line. The function of needle stone is to open the contact and create space for tapered with round **ONLY!**
- Keep your hand parallel to the long axis **at all times!**



When prepping a proximal surface, the next logical question to ask is :

How to prevent injury to the adjacent tooth?

Injury can be prevented by these methods:

- Creating an **Enamel Lip** to act as a shield for the adjacent tooth.
- Using a matrix band to protect the adjacent tooth while opening the contact. .



Proximal Reduction



Finishing your Prep





We promise you after this step, your’e really done!

Let’s talk about finishing your prep.
This step is often skipped by students thinking that it does not matter.
Let us tell you why it **does** matter:

- Blend the palatal finish line with the buccal finish line smoothly.
- Removing any sharp /undermined enamel.
- Removing irregularities provides better retention of the crown.
- Improve marginal fit of the crown.
- Reduce stress concentration.



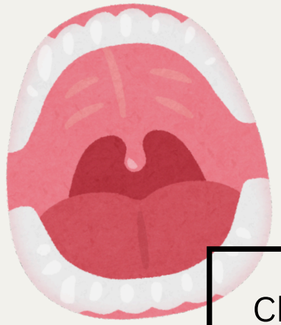
Conclusion: Do **NOT** skip finishing if you want that A+ Prep!





Finishing your Prep

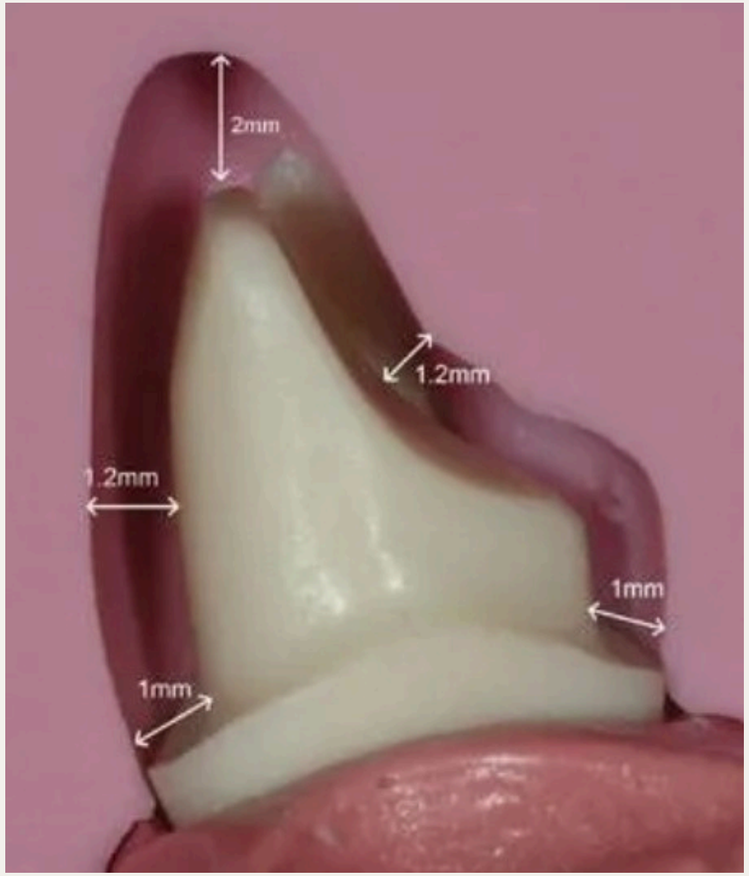




How to check your work

Follow this list to check your work:

Check the AMOUNT of reduction done using index and probe	Check that the borders of the lingual fossa are preserved not lost	check that there is no roughness or any sharp angles
Check that finish line is continous, all on the same level and has no irregularities	check that your taper is correct -no undercuts/over convergence	
Check that lingual fossa is properly reduced	Check that both buccal and palatal finish lines blend together smoothly	
check that there is no injury to adjacent tooth	Check that the cingilum is preserved and not lost	
check that the contact is open properly	check that secondary plane is done properly	





How to check your work



Congrats! You have now mastered All Ceramic crown preps!





Contact Us!



If you have any further questions, stuck in any step or want to show us your progress, feel free to contact us completely free of charge!

Our talented instructor Dr. Salma is waiting for you!

Want to book a one on one session? Just give us a DM!
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https://www.tiktok.com/@dentahelp?_r=1&_t=ZS-97oeLMuArIr
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Thank You





